Sentiment Analysis of Bengali Book Reviews for Literary Insights

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***Abstract*—The rise of Bengali literature and online book review platforms reflects readers opinions and emotions.However, analyzing sentiment in Bengali text is challenging because of the language's complex structure and the lack of high-quality datasets. This study focuses on sentiment analysis of Bengali book reviews using transformer-based models, including BanglaBERT, mBERT (Cased and Uncased), and XLM-RoBERTa. We collected 110,645 Bengali book reviews and reduced them to 21,446 samples through preprocessing and class balancing to improve model performance. The results show that mBERT (Uncased) and BanglaBERT performed the best in accurately identifying sentiment in Bengali book reviews. This research proves that transformer-based models are highly effective for improving sentiment classification in Bengali text, helping to better understand reader feedback.**

***Index Terms*—Bengali book review, NLP, BanglaBERT, mBERT cased, mBERT uncased, XLM-Roberta, Text classification,Transformer.**

I. Introduction

The increased use of social media platforms and online platforms has gotten more eyes on sentiment writing in the Bengali language. Analysis of such sentiments in book reviews in the Bengali language faces several challenges: their complex linguistic structure, limited availability of large-scale datasets, and the subtle expression of things in typical literary reviews.

In this study, we focus on the sentiment analysis of Bengali book reviews using transformer-based models like BanglaBERT, mBERT (Cased and Uncased), and XLM-RoBERTa. The final balanced dataset features 21,446 samples after class balancing and augmentation for the initial dataset of 110,645 Bengali book reviews collected and preprocessed.

Our findings indicate that mBERT Uncased has produced the best accuracy score of 85.69%, with BanglaBERT and mBERT Cased not far behind with high precision and recall scores. From this, the merit of using transformer models to capture sentiment nuances in Bengali text for enhancing sentiment classification of an under-resourced language has thus been demonstrated.

The rest of the paper is structured as follows: Section II is a literature review on book review for the Bengali language. Section III describes the dataset, the preprocessing steps, and data visualization. Section IV includes the methodology,and transformerbased models. Section V discusses the results and compares the performance of different models. Finally, Section VI is the conclusion.

II. Literature Review

[1]**BanglaBook: A Large-scale Bangla Dataset for Sentiment Analysis from Book Reviews** The article presents the BanglaBook dataset, which is a large dataset of book reviews in Bengali. It is used to train various sentiment classifying models like Logistic Regression (LR), Support Vector Machine (SVM), Naïve Bayes, LSTM, and Transformer-based BERT, etc. The best accuracy of 88.5% is obtained by BanglaBERT.

[2]**A Systematic Review of Sentiment Analysis from Bengali Text using NLP** This paper is all about different approaches to work in Bengali text for Implementing sentiment analysis. BARD, Bengali IMDb Reviews, and BanglaBook are tried by authors. TF-IDF, Word2Vec embeddings, BiLSTM, and GRU are techniques that are examined. As a matter of fact, transformer-based models are better than machine learning methods each time.

[3]**Book Review Sentiment Classification in Bangla using Deep Learning and Transformer Model** This paper presents a dataset of 5,189 Bangla book reviews crawled from

Rokomari.com, manually annotated into positive and negative sentiments. It is used to train various sentiment classification models like LSTM, BiLSTM, CNN-LSTM, CNN-BiLSTM, and Transformer-based models like m-BERT, Bangla-BERT, and XLM-R. XLM-R showed the highest F1-score of 88.95%.

[4]**Classification of Book Review Sentiment in Bangla Language Using NLP, Machine Learning and LSTM** This paper also presents a dataset of 6,281 Bangla book reviews scraped from Facebook groups and online bookstores. This is used to train various sentiment classification models like Logistic Regression, SVM, Decision Tree,KNN, Random Forest, and LSTM. It achieved the best accuracy of 97.49% with the LSTM model, outperforming traditional machine learning approaches.

[5]**Sentiment Analysis of Book Review in Bangla Using NLP and Machine Learning** The research collected a dataset of approximately 5,500 Bangla comments from various book-selling websites and Facebook Bangla book reviews. The compared models are AdaBoost, Decision Tree (DT), Random Forest Tree (RFT) , SVM, and LightGBM. Among them, Random Forest Tree provides the maximum accuracy of 99.39%.

[6]**Sentiment Polarity Detection on Bengali Book Reviews Using Multinomial Naïve Bayes** The article proposes a sentiment polarity detection system for Bengali book reviews based on a corpus of 2000 labeled reviews. The approach, with the Multinomial Naive Bayes algorithm and unigram features, achieves 84% accuracy on the test set, being effective in classifying sentiments as positive or negative.

[7]**A Machine Learning Approach for Sentiment Analysis of Book Reviews in Bangla** In this paper, a dataset of 10,000 Bengali book reviews is analyzed. The models tried out DT, RF , and CNN. The best accuracy of 85%. is achieved by Random Forest among them.

III. Data Analysis

The dataset used for this research consists of 110,645 Bengali book reviews. It is a multimodal dataset with three sentiment classes which is positive, negative, and neutral. The distribution of sentiment labels is highly imbalanced with 99,110 positive, 6,772 negative class, and 4,763 neutral samples. The dataset have multiple attributes like ID, book name, writer name, category, rating, review text, site, sentiment, and label. However,the most relevant attributes are retained for sentiment classification . We keep the rating attribute which ranges from 1 to 5, the review section which is the textual content used for classification, the sentiment attribute which is a categorical class with values of positive, negative, or neutral, and the label attribute which represents sentiment numerically. Figure 1 shows the dataset.

To handle class imbalance we applied resample to the positive class and reduced it to 10% of its original size. The final dataset after removing majority of the positive class contains 21,446 total samples, with 9,911 positive, 6,772 negative class, and 4,763 neutral samples. Figure 2 shows the dataset Structure.

	id	Book_Name	Writer_Name	Category	Rating	Review	Site	sentiment	label
91566	129641	অন্তু (হার্ডকভার)	শরীফুল হাসান	থ্রিলার	3	ব্যক্তিগতভাবে আমার ভাল লেগেছে বইটা অতিলৌকিক ক...	Rokomari	neutral	1
34245	102022	প্রোডাক্টিভ মুসলিম (হার্ডকভার)	মোহাম্মদ ফারিস	ইসলামি বই, আত্ম-উন্নয়ন	4	সুন্দর লেখা, ভালো লাগলো	Rokomari	positive	2
86126	124395	রকমারি কালেকশন অফ রবীন্দ্রনাথ ঠাকুর (৪টি বই)	রবীন্দ্রনাথ ঠাকুর	চিরায়ত উপন্যাস	2	বইঃ মৌকাতুলিলেখক রবীন্দ্রনাথ ঠাকুরঃরমেশের যোর...	Rokomari	negative	0
104639	92946	পরির হাতে নীল চুড়ি	মমীন মুহাসিন	শিশু-কিশোর গল্প	5	অসাধারণ গল্প। বেশ ভালো লেগেছে।	Rokomari	positive	2
17672	64381	রাহে বেলায়াত (হার্ডকভার) আল্লামা ইবনে কাইয়্যিম রহ...	ড. খোন্দকার আব্দুল্লাহ জাহাঙ্গীর	আত্মোন্নতি ও সুফিবাদ	5	প্রতিটি মুসলিমের ঘরে এই বইটা থাকা উচিত	Rokomari	positive	2

Fig. 1: Overview of the dataset

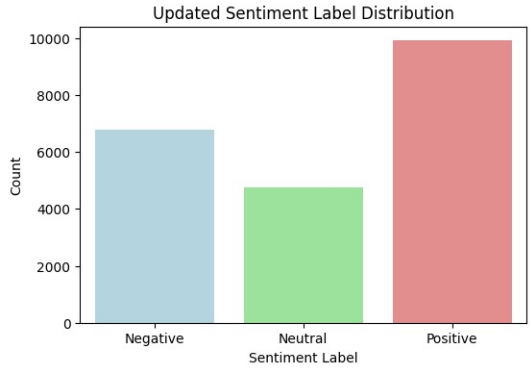

Fig. 2: Overview of the dataset Structure

Several preprocessing steps are applied to clean and prepare the dataset. Initially, the preprocessing steps are designed to support both traditional machine learning models and transformer models, but after exploration, we decided to focus solely on transformer-based models as they are better at handling textual data. The review section contained English words, emojis, and numbers, which are removed to keep the data clean. Only Bengali characters were retained, and extra spaces were eliminated. For tokenization, we use Bangla BERT tokenizer, which performs subword tokenization. Unlike traditional machine learning models, stopwords are necessary for transformers to understand context, so they are not removed. For transformer-based models, Bangla BERT tokenizer use with a maximum sequence length of 128, with padding and truncation enabled. The tokenized text is then converted into PyTorch tensors, making it suitable for deep learning models.

IV. Methodology

A. Overview

This research focuses on using transformer-based models to classify Bengali book reviews into positive, neutral, and negative sentiments. Since traditional machine learning models were not used the study primarily explores pretrained transformer models to achieve better sentiment classification.

After preprocessing as discussed in the data analysis section we selected transformer-based models to classify Bengali Book Review. The models were trained, tuned, and validated on a preprocessed dataset.

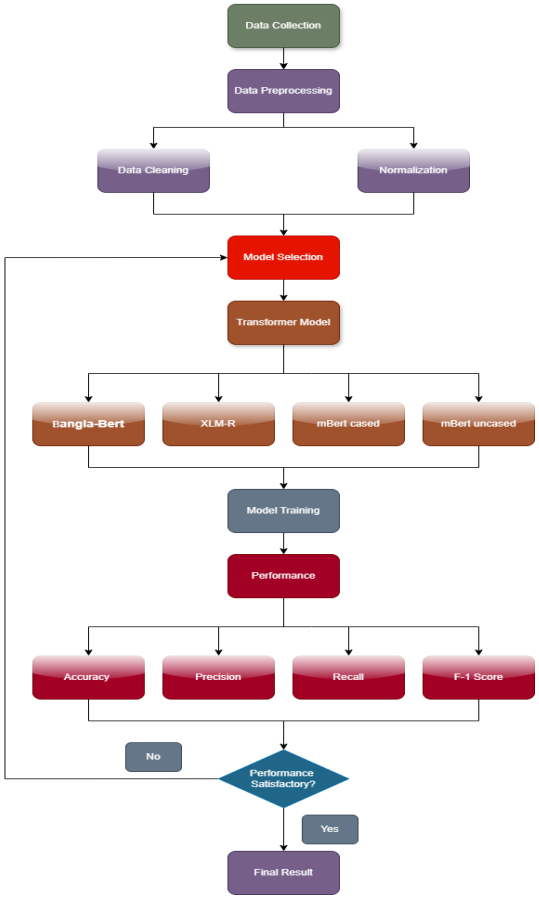

Fig. 3: Overall workflow

The overall workflow Figure 3.

B. Transformer-Based Models

To determine the most effective model for Bengali sentiment classification we used four pretrained transformer models.

1) **BanglaBERT:** BanglaBERT is a Bengali-specific transformer model trained on a large-scale Bengali corpus.It is fine-tuned to understand the language-specific features of Bengali text more effectively.
2) **Multilingual BERT (mBERT):** mBERT is a multilingual transformer model trained on 104 languages which includes Bengali also. Both cased and uncased versions were used to evaluate their effectiveness.
   Cased Version: Retains capitalizations for better differentiation. But since Bengali is not a case sensitive language its not properly suitable. But we selected it because it uses tokenization differently than Uncased version of mBERT which might be better.
   Uncased Version : Normalizes case sensitivity improving tokenization in some cases.
3) **XLM-RoBERTa (XLM-R):** XLM-RoBERTa is a robust multilingual transformer model optimized for cross-lingual understanding. Capable of handling code-mixed Bengali text but performance varied due to different tokenization mechanisms.It usually does not work well for Bengali text than BanglaBERT and mBERT models.

C. Feature Selection

The input features included

1) **Input Features:**
   - **Rating (Numeric):**A numerical value representing user sentiment.The value is from 1 to 5.It was normalized.
   - **Review (Text):** The actual review content tokenized for the model.
2) **Target Variable (Label):** Sentiment Class (0, 1, 2):
   - 0 $\rightarrow$ Negative
   - 1 $\rightarrow$ Neutral
   - 2 $\rightarrow$ Positive

D. Data Splitting

To ensure a well-balanced training process, the dataset was split into training and testing.80% of the data was for training and 20% of the data was for testing and we also used stratified sampling. This approach ensures that each sentiment class (positive, neutral, and negative) is proportionally represented in both the training and testing datasets.

E. Hyperparameter Tuning for Transformer Models

We fine tuned several hyperparameters for training transformer models to get better performance from them:

- **Batch Size:** 16 - Balances memory usage and efficiency. We also tried batch size 8.
- **Learning Rate:** 2e-5 - Stable fine-tuning without rapid convergence issues. For more stable training we also used 1e-5. Initially used 1e-5 as it is useful for pretrained transformer models.
- **Weight Decay:** 0.01 - Prevents overfitting by reducing excessive updates. We tried 0.02 too to increase the regularization to prevent overfitting.
- **Training Epochs:** 5 - Higher epochs did not show significant improvement. We tried 6,7 epochs too. Evaluation Strategy: Epoch-wise validation for which performance assessed after each epoch.

F. Training and Evaluation Process

After defining the dataset and tuning hyperparameters, the transformer models were trained and evaluated using Hugging Face's Trainer API.

1) **Training Setup:**
   - The models were trained using PyTorch and Hugging Face's Trainer class, leveraging GPU acceleration.
   - Training logs were saved, and the best model was automatically loaded at the end of training.
2) **Evaluation Metrics:**

After training, we evaluated the models using multiple performance metrics like Accuracy,Precision,Recall and F1-Score and plotted confusion Matrix.

V. Result Analysis

We first applied transformer-based models like Bangla BERT, mBERT (both cased and uncased), and XLM-RoBERTa on the preprocessed dataset.

Model	Accuracy	Precision	Recall	F1-score
Bangla BERT	68.93%	0.68	0.69	0.68
mBERT Uncased	67.62%	0.67	0.68	0.67
mBERT Cased	67.02%	0.67	0.67	0.66
XLM-R	65.38%	0.63	0.65	0.63

TABLE I: Performance comparison of models on preprocessed dataset

The models performed poorly, particularly on the neutral class, where both precision and recall were significantly low. The worst performing model in the initial run was XLM-RoBERTa which achieved only 65.38% accuracy which is very low score for a transformer model. For the neutral class recall was 0.22 and F1-score was 0.30 indicating that it was very underperforming. The best-performing model in the initial run was Bangla BERT because it is specifically designed for Bengali text but even then it only achieved 68.93% accuracy, which is still low. It had better recall and F1-score compared to XLM-RoBERTa, but the results were still not satisfactory. Both variants of mBERT performed similarly showing no significant advantage over one another.

Figure 4 Shows the Confusion Matrix of BanglaBERT

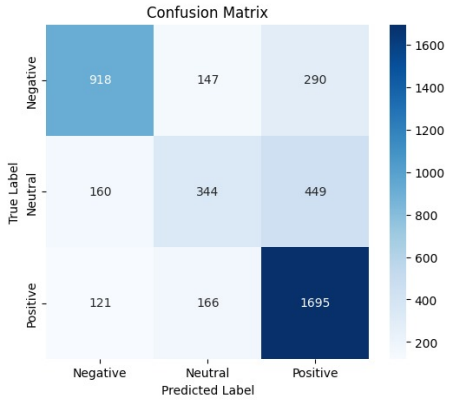

Fig. 4: Confusion Matrix of BanglaBERT

From the confusion matrix of Bangla BERT we noticed that it performed decently on positive and negative classes but misclassified many neutral reviews as positive which is not desirable. This same issue was observed across all the models in their confusion matrix as well.

One of the primary reasons for this poor performance is the nature of the dataset we worked with even after preprocessing. Most reviews in the dataset are only one or two words long and BERT-based models struggle with short text because they require more context to understand meaning. Neutral sentiment is already difficult to distinguish and the lack of sufficient words in reviews made it even harder for the model to classify them correctly. Although we tried weighted loss adjustments to handle the imbalanced data the impact was not sufficient.

Additionally we experimented further and removed the rating feature which is a numeric feature and training the models using only the review text but this actually worsened the results showing that the rating provided some valuable context for sentiment classification.

To improve classification and achieve better performance we applied additional data augmentation and class balancing techniques. Since short reviews were a major issue, we expanded single-word and two-word reviews by adding sentimentally relevant words based on their review classification. Positive reviews received positive reinforcement words, negative reviews received negative words, and the same was done for neutral ones. We were careful not to add duplicate words in the same review to prevent further issues.We also shuffled the dataset randomly to improve it even further

Even after augmentation the model did not improve significantly. So we took a further step to handle the class imbalance problem. We removed some data from the positive class as it was majority class and had more than twice as much data as neutral class and added more negative and neutral samples with similar contextual meanings. This was done carefully to prevent excessive repetition, as that could negatively impact the model's ability to generalize. After these preprocessing improvements, the models improved significantly.

Model	Accuracy	Precision	Recall	F1-score
mBERT Uncased	85.69%	0.8573	0.8569	0.8570
Bangla BERT	85.43%	0.8551	0.8543	0.8546
mBERT Cased	84.25%	0.8443	0.8425	0.8429
XLM-R	69.98%	0.7471	0.6998	0.6827

TABLE II: Performance comparison of models after further preprocessing

Here mBERT Uncased achieved the best performance with an accuracy of 85.69% and a precision of 0.8573, while mBERT Cased (84.25%) and Bangla BERT (85.43%) also performed well. Their precision was 0.8443 and 0.8551. The performance of the neutral sentiment class improved significantly—while still lower than the positive and negative classes it performed far better than before. The other evaluation metrics were also good this time.

Figure 5 Shows the Confusion Matrix of mBERT Uncased

From the confusion matrix of mBERT Uncased, it was clear that it performed well on all three sentiment classes, with its best performance on the negative class.

Even after these improvements XLM-RoBERTa still performed poorly. Although its accuracy improved compared to the first run but the recall and F1-score for the neutral class remained too low. We tried different hyperparameters to optimize XLM-RoBERTa, but it still underperformed. One possible reason is that it might not break down Bengali words as effectively as the other models. The results show that

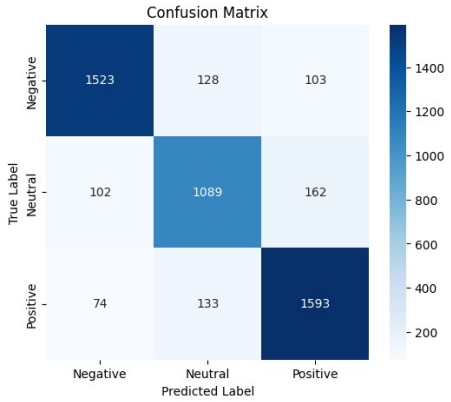

Fig. 5: Confusion Matrix of mBERT Uncased

XLM-RoBERTa is biased towards stronger sentiment classes (positive and negative) and does not work well for neutral reviews.

VI. Conclusion and Future Work

This study compared the performance of transformer-based models on pure sentiment analysis of Bengali book reviews. The results testify that mBERT (Uncased) and BanglaBERT were the winners with highest accuracy. XLM-RoBERTa also performed some improvement but was poor for neutral sentiment classification. The results thus testify that both language-specific and multilingual models can effectively tackle the challenging issues involved in Bengali sentiment classification. The suggested transformer models, as seen in the experiments, can enhance sentiment classification for low-resource languages like Bengali.

For Future work :

- **Dataset quality improvement:**Scaling of the dataset so that many more new book reviews can be added to gather different genres with different styles of writing to improve model generalization
- **Transformer model fine-tuning:**Further hyperparameter optimization and exploring advanced fine-tuning techniques to increase accuracy and convergence speed.

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